# Solution 1 Centered Gaussian Processes

**Problem 1** — Let $(X_t)_{t\in[0,1]}$ be a centered Gaussian process. We assume that the mapping $(t,\omega)\mapsto X_t(\omega)$ from $[0,1]\times\Omega$ into $\mathbb{R}$ is measurable. We denote the covariance function of $X$ by $K$.

**Problem 1.1** — Show that the mapping $t\mapsto X_t$ from $[0,1]$ into $L^2(\Omega)$ is continuous if and only if $K$ is continuous on $[0,1]^2$. In what follows, we assume that this condition holds.

We first assume that $K$ is continuous on $[0,1]^2$. Fix $t\in[0,1]$ and $\varepsilon>0$. By continuity at $(t,t)\in[0,1]^2$, we find that there exists a $\delta>0$ such that for all $u,v\in[t-\delta',t+\delta']$, we have $|K(u,v)-K(t,t)|<\varepsilon^2/3$. Then, for all $s\in[0,1]$ with $|t-s|<\delta$,

$$\mathbb{E}[(X_t-X_s)^2]=K(t,t)-2K(t,s)+K(s,s)<0+2\varepsilon^2/3+\varepsilon^2/3=\varepsilon^2.$$

thus

$$\|X_t-X_s\|_2<\varepsilon.$$

Conversely, assume that $t\mapsto X_t$ is continuous. Then, $t\mapsto X_t\mapsto\|X_t\|^2$ is continuous. $[0,1]$ is compact, thus $\|X_t\|^2$ attains some maximum value $M$. Fix $(t_1,t_2)\in[0,1]^2$ and $\varepsilon>0$. By continuity at $t_1$, we find a $\delta_1>0$ such that for all $s\in[t_1-\delta_1,t_1+\delta_1]$, we have $\|X_{t_1}-X_s\|_2<\frac{\varepsilon}{2M}$. Define $\delta_2$ similarly. Then, for all $s_1\in[t_1-\delta_1,t_1+\delta_1]$ and $s_2\in[t_2-\delta_2,t_2+\delta_2]$,

$$\begin{aligned}
|K(s_1,s_2)-K(t_1,t_2)| &= |\mathbb{E}[X_{s_1}X_{s_2}-X_{t_1}X_{t_2}]| \\
&\le |\mathbb{E}[X_{s_1}(X_{s_2}-X_{t_2})]|+|\mathbb{E}[(X_{s_1}-X_{t_1})X_{t_2}]| && \text{(Triangle Inequality)}\\
&\le \|X_{s_1}\|\,\|X_{s_2}-X_{t_2}\|+\|X_{s_1}-X_{t_1}\|\,\|X_{t_2}\| && \text{(Cauchy-Schwarz)}\\
&< M\frac{\varepsilon}{2M}+\frac{\varepsilon}{2M}M=\varepsilon.
\end{aligned}$$

as desired.

**Problem 1.2** — Let $h\colon[0,1]\to\mathbb{R}$ be a measurable function such that $\int_0^1|h(t)|\sqrt{K(t,t)}\,dt<\infty$. Show that, for a.e. $\omega$, the integral $\int_0^1 h(t)X_t(\omega)\,dt$ is absolutely convergent. We set $Z=\int_0^1 h(t)X_t\,dt$.

**Idea** (Key Ideas) — A continuous function on a compact domain is uniformly continuous.

We can take the measurable function $[0,1]\times\Omega\to\mathbb{R}$ seriously.

It is well-known (and otherwise quickly computed) that

$$\int_\Omega|X_t(\omega)|\,d\mathbb{P}=\sqrt{\frac{2}{\pi}}\sqrt{K(t,t)}.$$

The constant term doesn't matter.

We simply compute

$$\int_\Omega\int_0^1|h(t)X_t(\omega)|\,dt\,d\mathbb{P}=\int_0^1\int_\Omega|h(t)X_t(\omega)|\,d\mathbb{P}\,dt \qquad \text{(Fubini's Theorem)}$$

$$= \int_0^1 |h(t)| \int_\Omega |X_t(\omega)| \, d\mathbb{P} \, dt$$
$$= \int_0^1 |h(t)| \sqrt{\frac{2}{\pi}} \sqrt{K(t,t)} \, dt < \infty.$$

Thus, for a.e. $\omega$, the integral $\int_0^1 h(t) X_t(\omega) \, dt$ is absolutely convergent.

**Problem 1.3 —** We now make the stronger assumption $\int_0^1 |h(t)| \, dt < \infty$. Show that $Z$ is the $L^2$-limit of the variables

$$Z_n = \sum_{i=1}^n X_{i/n} \int_{(i-1)/n}^{i/n} h(t) \, dt$$

when $n \to \infty$ and infer that $Z$ is a Gaussian random variable.

Let $M = \int_0^1 |h(t)| \, dt$. Let's go ahead and compute the difference:

$$\begin{aligned} Z_n - Z &= \sum_{i=1}^n X_{i/n} \int_{(i-1)/n}^{i/n} h(t) \, dt - \int_0^1 h(t) X_t \, dt \\ &= \sum_{i=1}^n X_{i/n} \int_{(i-1)/n}^{i/n} h(t) \, dt - \sum_{i=1}^n \int_{(i-1)/n}^{i/n} h(t) X_t \, dt \\ &= \sum_{i=1}^n \int_{(i-1)/n}^{i/n} h(t)(X_{i/n} - X_t) \, dt \end{aligned}$$

We now bound the norm of the whole thing, using triangle inequality.

$$\|Z_n - Z\|_2 \le \sum_{i=1}^n \int_{(i-1)/n}^{i/n} h(t) \left\|(X_{i/n} - X_t)\right\|_2 \, dt$$

By uniform continuity of $t \mapsto X_t$, for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $s \in [t-\delta, t+\delta]$, we have $\|X_t - X_s\|_2 < \varepsilon/M$. Then pick some $n > \frac{1}{\delta}$. We have

$$\begin{aligned} &\le \sum_{i=1}^n \int_{(i-1)/n}^{i/n} h(t) \varepsilon/M \, dt \\ &= \varepsilon/M \int_0^1 |h(t)| \, dt \\ &= \varepsilon. \end{aligned}$$

Thus, $Z_n \to Z$ in $L^2$. The limit of a $L^2$-convergent sequence of Gaussian random variables is a Gaussian random variable.

**Problem 1.4 —** We assume that $K$ is twice continuously differentiable. Show that, for every $t \in [0,1]$, the limit $\dot{X}_t = \lim_{s \to t} \frac{X_s - X_t}{s-t}$ exists in $L^2(\Omega)$. Verify that $(\dot{X}_t)_{t \in [0,1]}$ is a centered Gaussian process and compute its covariance function.

Fix $t \in [0,1]$. For the first part, the existence of continuous second derivatives implies that $\exists \partial_{xy} K \forall \varepsilon > 0 \exists \delta_\varepsilon > 0$ such that $\forall |\alpha|, |\beta| < \delta_\varepsilon$,

$$\left| \frac{K(t+\alpha, t+\beta) - K(t+\alpha, t) - K(t, t+\beta) + K(t,t)}{\alpha\beta} - \partial_{xy} K(t,t) \right| < \varepsilon.$$

Fix $t \in [0,1]$. We will show that $\dot{X}_t$ exists by showing that the sequence $Y_\alpha = \frac{X_{t+\alpha} - X_t}{\alpha}$ is Cauchy in $L^2$.

**Idea** (Cauchy Criterion) — For any $\varepsilon > 0$, I will exhibit a $|\delta| > 0$ such that for all $|\delta| > |\eta| > 0$, $\|Y_\delta - Y_\eta\|_2 < \varepsilon$.

Pick any $\varepsilon$. Now, we will pick $|\delta| = \delta_{\varepsilon/4}$ (where $\delta$ is defined as above). For any $|\eta| < |\delta|$, we have

$$\begin{aligned}
\left\| \frac{X_{t+\delta} - X_t}{\delta} - \frac{X_{t+\eta} - X_t}{\eta} \right\|_2^2 &= \mathbb{E}\left[ \left( \frac{X_{t+\delta} - X_t}{\delta} - \frac{X_{t+\eta} - X_t}{\eta} \right)^2 \right] \\
&= \left| \frac{K(t+\delta, t+\delta) - 2K(t+\delta, t) + K(t,t)}{\delta^2} \right. \\
&+ \frac{K(t+\eta, t+\eta) - 2K(t+\eta, t) + K(t,t)}{\eta^2} \\
&\left. -2\frac{K(t+\delta, t+\eta) - K(t+\delta, t) - K(t, t+\eta) + K(t,t)}{\delta\eta} \right| \\
&< \varepsilon
\end{aligned}$$

as desired.

We need to show that $\dot{X}_t$ is a centered Gaussian *process* i.e. any finite linear combination of $\dot{X}_t$'s is a centered Gaussian RV. Let's investigate $\sum_i \alpha_i \dot{X}_i$ for some finite set of indices $i \in I \subset [0,1]$. By (finite) linearity of limits, this is equal to the $L^2$-limit of $\sum_i \alpha_i Y_{i,\delta}$ as $\delta \to 0$... but this is just a $L^2$-limit of a bunch of centered Gaussian RVs, so it's not too deep. Let's compute the variance of this process in the case where $I = \{t, s\}$:

$$\begin{aligned}
\mathbb{E}\left[ (\alpha_t Y_{t,\delta} + \alpha_s Y_{s,\delta})^2 \right] &= \mathbb{E}\left[ \alpha_t^2 Y_{t,\delta}^2 + 2\alpha_t\alpha_s Y_{t,\delta} Y_{s,\delta} + \alpha_s^2 Y_{s,\delta}^2 \right] \\
&= \alpha_t^2 \left( \frac{K(t+\delta, t+\delta) - 2K(t+\delta, t) + K(t,t)}{\delta^2} \right) \\
&+ 2\alpha_t\alpha_s \left( \frac{K(t+\delta, s+\delta) - K(t+\delta, s) - K(t, s+\delta) + K(t,s)}{\delta^2} \right) \\
&+ \alpha_s^2 \left( \frac{K(s+\delta, s+\delta) - 2K(s+\delta, s) + K(s,s)}{\delta^2} \right)
\end{aligned}$$

thus in the limit $\alpha_t X_t + \alpha_s X_s$ (from that same result on convergence of Gaussian RVs) is a centered Gaussian RV with variance

$$\alpha_t^2 \partial_{xy} K(t,t) + 2\alpha_t\alpha_s \partial_{xy} K(t,s) + \alpha_s^2 \partial_{xy} K(s,s).$$

This immediately shows that the covariance function of $\dot{X}$ is $\partial_{xy} K$ (by e.g. taking the results for $(\alpha_t, \alpha_s) = (1,0), (0,1), (1,1)$ and subtracting).

# Solution 2 Kalman Filtering

**Problem 2 —** Let $(\varepsilon_n)_{n\geq 0}$ and $(\eta)_{n\geq 0}$ be two independent sequences of independent Gaussian random variables such that, for every $n$, $\varepsilon_n$ is distributed according to $\mathcal{N}(0,\sigma^2)$ and $\eta_n$ is distributed according to $\mathcal{N}(0,\delta^2)$, where $\sigma > 0$ and $\delta > 0$. We consider two other sequences $(X_n)_{n\geq 0}$ and $(Y_n)_{n\geq 0}$ defined by the properties $X_0 = 0$, and, for every $n \geq 0$, $X_{n+1} = a_n X_n + \varepsilon_{n+1}$ and $Y_n = cX_n + \eta_n$, where $c$ and $a_n$ are positive constants. We set

$$\hat{X}_{n|n} = \mathbb{E}[X_n | Y_0, Y_1, \ldots, Y_n]$$

and

$$\hat{X}_{n+1|n} = \mathbb{E}[X_{n+1} | Y_0, Y_1, \ldots, Y_n].$$

The goal of the exercise is to find a recursive formula allowing one to compute these conditional expectations.

**Problem 2.1 —** Verify that $\hat{X}_{n+1|n} = a_n \hat{X}_{n|n}$, for every $n \geq 0$.

This is clear:

$$\begin{aligned}\hat{X}_{n+1|n} &= \mathbb{E}[a_n X_n + \varepsilon_{n+1} | Y_0, Y_1, \ldots, Y_n] \\ &= a_n \mathbb{E}[X_n | Y_0, Y_1, \ldots, Y_n] + \underbrace{\mathbb{E}[\varepsilon_{n+1} | Y_0, Y_1, \ldots, Y_n]}_{\star} \\ &= a_n \hat{X}_{n|n}.\end{aligned}$$

where $\star$ vanishes because we are conditioning on some function of $\{\varepsilon_i\}_{i=0}^n$ and $\{\eta_i\}_{i=0}^n$, which are all independent of $\varepsilon_{n+1}$.

**Problem 2.2 —** Show that, for every $n \geq 1$,

$$\hat{X}_{n|n} = \hat{X}_{n|n-1} + \frac{\mathbb{E}[X_n Z_n]}{\mathbb{E}[Z_n^2]} Z_n,$$

where $Z_n = Y_n - c\hat{X}_{n-1|n-1}$.

**Idea —** This problem, rephrased geometrically, is the following: I have a subspace $U \subset V$. I have a vector $X \in V$; it has already been projected onto $U$, $p_U X$. Now, I want to create $U'$ as the span of $U$ and another vector $Y$. I want to investigate $p_{U'} X$. To do this, I first take the component of $Y$ which is orthogonal to $U$ so I can work with two orthogonal subspaces; let $Z = Y - p_U Y$. Then, $p_{U'} X = p_U X + p_Z(X)$. Let's do this for real now.

Let $V_n = \operatorname{span}(Y_0, \ldots, Y_n)$.

We know that $\hat{X}_{n|n}$ is the orthogonal projection of $X_n$ onto $V_n$; we just need to verify that the RHS is as well. The RHS clearly lies in $V_n$, because $\hat{X}_{n|n-1}$ is an orthogonal projection into $V_{n-1}$, and $Z_n$ is a linear combination of $Y_n$ and an orthogonal projection into $V_{n-1}$.

Now, we need to verify that $X_n$ minus the RHS is orthogonal to $V_n$:

$$\star = \underbrace{(X_n - \hat{X}_{n|n-1})}_{\in V_{n-1}^{\perp}} - \frac{\mathbb{E}[X_n Z_n]}{\mathbb{E}[Z_n^2]} Z_n$$

The first component is orthogonal to $V_{n-1}$ by definition of the orthogonal projection. Similarly,

$$Z_n = Y_n - c\hat{X}_{n-1|n-1} = c\underbrace{\left(X_{n-1} - \hat{X}_{n-1|n-1}\right)}_{\in V_{n-1}^{\perp}} + \eta_n \in V_{n-1}^{\perp}$$

thus $\star$ is orthogonal to $V_{n-1}$.

Alright, so we just need to show that $\star$ is orthogonal to $Y_n = cX_n + \eta_n$. We already know $\star$ is orthogonal to $c\hat{X}_{n-1|n-1}$, so we just need to show that $\star$ is orthogonal to $Z_n$. First off, let's compute this trick:

$$\mathbb{E}[\hat{X}_{n|n-1}\left(X_n - \hat{X}_{n|n-1}\right)] = \mathbb{E}[\mathbb{E}[\hat{X}_{n|n-1}\left(X_n - \hat{X}_{n|n-1}\right)|Y_0, \ldots, Y_{n-1}]] = 0$$

We compute:

$$\begin{aligned}
\mathbb{E}[\star Z_n] &= \mathbb{E}[X_n Z_n] - \mathbb{E}[\hat{X}_{n|n-1} Z_n] - \frac{\mathbb{E}[X_n Z_n]\mathbb{E}[Z_n^2]}{\mathbb{E}[Z_n^2]} \\
&= -\mathbb{E}[\hat{X}_{n|n-1} Z_n] \\
&= -\mathbb{E}[\hat{X}_{n|n-1}\eta_n] - c\underbrace{\mathbb{E}[\hat{X}_{n|n-1}\left(X_n - \hat{X}_{n|n-1}\right)]}_{0} \\
&= 0
\end{aligned}$$

and we are done.

**Problem 2.3 —** Evaluate $\mathbb{E}[X_n Z_n]$ and $\mathbb{E}[Z_n^2]$ in terms of $P_n = \mathbb{E}[(X_n - X_{n|n-1})^2]$ and infer that, for every $n \geq 1$,

$$\hat{X}_{n+1|n} = a_n\left(\hat{X}_{n|n-1} + \frac{cP_n}{c^2 P_n + \delta^2} Z_n\right).$$

Sure. First, we compute

$$\begin{aligned}
\mathbb{E}[X_n Z_n] &= \mathbb{E}[X_n\left(c(X_n - \hat{X}_{n|n-1}) + \eta_n\right)] \\
&= c\mathbb{E}[X_n\left(X_n - \hat{X}_{n|n-1}\right)] \\
&= c\mathbb{E}[X_n\left(X_n - \hat{X}_{n|n-1}\right)] + \underbrace{\mathbb{E}[\hat{X}_{n|n-1}\left(X_n - \hat{X}_{n|n-1}\right)]}_{0} \\
&= c\mathbb{E}[(X_n - \hat{X}_{n|n-1})^2] = cP_n.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{E}[Z_n^2] &= \mathbb{E}[\left(c(X_n - \hat{X}_{n|n-1}) + \eta_n\right)^2] \\
&= c^2\mathbb{E}[(X_n - \hat{X}_{n-1|n-1})^2] + \mathbb{E}[\eta_n^2] \\
&= c^2 P_n + \delta^2.
\end{aligned}$$

Thus, it is immediate that

$$\hat{X}_{n|n} = \hat{X}_{n|n-1} + \frac{cP_n}{c^2 P_n + \delta^2} Z_n.$$

The result follows from combining with part (1).

**Problem 2.4 —** Verify that $P_1 = \sigma^2$ and that, for every $n \geq 1$, the following induction formula holds:

$$P_{n+1} = \sigma^2 + a_n^2 \frac{\delta^2 P_n}{c^2 P_n + \delta^2}.$$

We compute

$$\begin{aligned} P_1 &= \mathbb{E}[(X_1 - X_{1|0})^2] \\ &= \mathbb{E}[(\varepsilon_1 - \underbrace{\mathbb{E}[\varepsilon_1 | \eta_0]}_{0})^2] \\ &= \sigma^2. \end{aligned}$$

Now let's take the final result of problem 2.3 and subtract both sides from $X_{n+1}$. The LHS is just $X_{n+1} - \hat{X}_{n+1|n}$, and the RHS is

$$\star = \varepsilon_{n+1} + a_n \left( X_n - \hat{X}_{n|n-1} - \frac{cP_n}{c^2 P_n + \delta^2} Z_n \right).$$

If we take the norm-squared of each side now, the LHS becomes $P_{n+1}$, and the RHS becomes

$$\mathbb{E}[\star^2] = \mathbb{E}[\varepsilon_{n+1}^2] + a_n^2 \mathbb{E}[\left( X_n - \hat{X}_{n|n-1} - \frac{cP_n}{c^2 P_n + \delta^2} Z_n \right)^2]$$

(the cross-term is zero because $\varepsilon_{n+1}$ is independent of the remaining terms),

$$\begin{aligned} &= \sigma^2 + a_n^2 \mathbb{E}[\left( X_n - \hat{X}_{n|n-1} \right)^2] \\ &- 2a_n^2 \frac{cP_n}{c^2 P_n + \delta^2} \mathbb{E}[\left( X_n - \hat{X}_{n|n-1} \right) Z_n] \\ &+ a_n^2 \frac{c^2 P_n}{c^2 P_n + \delta^2} \mathbb{E}[Z_n^2] \\ &= \sigma^2 + a_n^2 P_n \\ &- 2a_n^2 \frac{cP_n}{c^2 P_n + \delta^2} \left( cP_n + \underbrace{\mathbb{E}\left[ (X_n - \hat{X}_{n|n-1}) \eta_n \right]}_{0} \right) \\ &+ a_n^2 \frac{c^2 P_n^2}{(c^2 P_n + \delta^2)^2} \left( c^2 P_n + \delta^2 \right) \\ &= \sigma^2 + a_n^2 \left( P_n - \frac{c^2 P_n^2}{c^2 P_n + \delta^2} \right) \\ &= \sigma^2 + a_n^2 \frac{\delta^2 P_n}{c^2 P_n + \delta^2}. \end{aligned}$$

Thus, we are done.

# Solution 3 Conditional Independence

**Problem 3 —** Let $H$ be a (centered) Gaussian space and let $H_1$ and $H_2$ be linear subspaces of $H$. Let $K$ be a closed linear subspace of $H$. We write $p_K$ for the orthogonal projection onto $K$. Show that the condition

$$\forall X_1 \in H_1, \quad \forall X_2 \in H_2, \quad \mathbb{E}[X_1 X_2] = \mathbb{E}[p_K(X_1) p_K(X_2)]$$

implies that the $\sigma$-fields $\sigma(H_1)$ and $\sigma(H_2)$ are conditionally independent given $\sigma(K)$. (This means that, for every nonnegative $\sigma(H_1)$-measurable random variable $X_1$, and for every nonnegative $\sigma(H_2)$-measurable random variable $X_2$, one has $\mathbb{E}[X_1 X_2 | K] = \mathbb{E}[X_1|K]\mathbb{E}[X_2|K]$.)

$\mathbb{E}[(X_1 - p_K(X_1))p_K(X_2)] = 0$, because by definition $p_K(X_2)$ lies inside $L^2(\Omega, \sigma(K), \mathbb{P})$, and $X_1 - p_K(X_1)$ should be orthogonal to it. Symmetrically, $\mathbb{E}[(X_2 - p_K(X_2))p_K(X_1)] = 0$. Thus, we have

$$\mathbb{E}[(X_1 - p_K(X_1))(X_2 - p_K(X_2))] = \mathbb{E}[X_1 X_2] - \mathbb{E}[p_K(X_1)p_K(X_2)] = 0$$

for all $X_1 \in H_1$ and $X_2 \in H_2$. This is clearly also true for all $X_1 \in \operatorname{span}(H_1, K)$ and $X_2 \in \operatorname{span}(H_2, K)$. Thus, without loss of generality, $K \subset H_1$ and $K \subset H_2$. $K$ is closed, and thus admits an orthogonal complement in $H_1$ and $H_2$. Thus, $H_1 = M_1 \oplus K$ and $H_2 = M_2 \oplus K$, where $M_1$ is orthogonal to $K$ and $M_2$ is orthogonal to $K$. Then, the statement above simply reads that $M_1$ is orthogonal to $M_2$. All of these are Gaussian spaces, thus $\sigma(M_1)$, $\sigma(M_2)$, and $\sigma(K)$ are all mutually (not merely pairwise!) independent.

**Idea —** Now, the remaining proof is "obvious." There are just lots of details, where we show it for boxes then bootstrap with monotone class.

Now, let me show that in the specific case where $X_1 = \mathbb{1}_{A_1}\mathbb{1}_B$ and $X_2 = \mathbb{1}_{A_2}\mathbb{1}_C$, where $A_1 \in \sigma(M_1)$, $A_2 \in \sigma(M_2)$, and $B, C \in \sigma(K)$, we have $\mathbb{E}[X_1 X_2|K] = \mathbb{E}[X_1|K]\mathbb{E}[X_2|K]$. I will do this by considering some element of $\sigma(K)$, $D \in \sigma(K)$. On the LHS, have

$$\mathbb{E}[\mathbb{1}_D X_1 X_2] = \mathbb{E}[\mathbb{1}_D \mathbb{1}_{A_1}\mathbb{1}_B \mathbb{1}_{A_2}\mathbb{1}_C] = \mathbb{P}(A_1 \cap A_2 \cap (B \cap C \cap D)) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(B \cap C \cap D)$$

where we split by mutual independence. On the RHS, we have to do some symbol-pushing. First off, $\mathbb{E}[X_2|K]$ is $\sigma(K)$-measurable, thus

$$\mathbb{E}[\mathbb{1}_D \mathbb{E}[X_1|K]\mathbb{E}[X_2|K]] = \mathbb{E}[\mathbb{1}_D \mathbb{E}[X_1 \mathbb{E}[X_2|K]|K]] = \mathbb{E}[\mathbb{1}_D X_1 \mathbb{E}[X_2|K]] = \mathbb{E}[\mathbb{1}_{A_1}\mathbb{1}_D \mathbb{1}_B \mathbb{E}[X_2|K]]$$

Now, $\mathbb{1}_{A_1}$ is $\sigma(M_1)$-measurable, while $\mathbb{1}_D \mathbb{1}_B \mathbb{E}[X_2|K]$ is $\sigma(K)$-measurable; thus they are independent, and we can split the expectation:

$$\begin{aligned}
\mathbb{E}[\mathbb{1}_{A_1}\mathbb{1}_D \mathbb{1}_B \mathbb{E}[X_2|K]] &= \mathbb{E}[\mathbb{1}_{A_1}]\mathbb{E}[\mathbb{1}_B \mathbb{1}_D \mathbb{E}[X_2|K]] \\
&= \mathbb{P}(A_1)\mathbb{E}[\mathbb{1}_B \mathbb{1}_D \mathbb{1}_{A_2}\mathbb{1}_C] \\
&= \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(B \cap C \cap D)
\end{aligned}$$

Great. We have thus shown that for all $X_1 = \mathbb{1}_{A_1}\mathbb{1}_B$ and $X_2 = \mathbb{1}_{A_2}\mathbb{1}_C$, where $A_1 \in \sigma(M_1)$, $A_2 \in \sigma(M_2)$, and $B, C \in \sigma(K)$, we have $\mathbb{E}[X_1 X_2|K] = \mathbb{E}[X_1|K]\mathbb{E}[X_2|K]$. Now, by linearity of expectation, the set of $\sigma(M_1, K)$-measurable functions $Y_1$ such that $\mathbb{E}[Y_1 X_2|K] = \mathbb{E}[Y_1|K]\mathbb{E}[X_2|K]$ for all $X_2$ of the above form is a vector space, and it contains the generating $\pi$-system $\{\mathbb{1}_{A_1}\mathbb{1}_B\}$. Thus, by monotone class, it actually just

contains all $\sigma(M_1, K)$-measurable functions. Similarly, the set of all $\sigma(M_2, K)$-measurable functions $Y_2$ such that $\mathbb{E}[Y_1 Y_2 | K] = \mathbb{E}[Y_1 | K]\mathbb{E}[Y_2 | K]$ for all $\sigma(M_1, K)$-measurable $Y_1$ is a vector space containing the generating $\pi$-system $\{\mathbb{1}_{A_2}\mathbb{1}_C\}$, and thus contains all $\sigma(M_2, K)$-measurable functions. Thus, $\sigma(M_1, K)$ and $\sigma(M_2, K)$ are conditionally independent given $\sigma(K)$.

# Solution 4 Lévy's Construction of Brownian Motion

**Problem 4** — For every $t \in [0,1]$, we set $h_0(t) = 1$, and then, for every integer $n \geq 0$ and every $k \in \{0, 1, \ldots, 2^n - 1\}$,

$$h_k^n(t) = 2^{n/2}\left(\mathbb{1}_{[(2k)2^{-n-1},(2k+1)2^{-n-1}]}(t) - \mathbb{1}_{[(2k+1)2^{-n-1},(2k+2)2^{-n-1}]}(t)\right).$$

**Problem 4.1** — Verify that the functions $h_0, (h_k^n)_{0\leq k\leq 2n-1}$ form an orthonormal basis of $L^2([0,1]; B([0,1]); dt)$.

**Idea** (Hint) — Observe that, for every fixed $n \geq 0$, any function $f\colon [0,1] \to \mathbb{R}$ that is constant on every interval of the form $[(j-1)2^{-n}, j2^{-n}]$, for $1 \leq j \leq 2^n$, is a linear combination of the functions $h_0, (h_k^m)_{0\leq k\leq 2^m-1}$.

First we verify that the functions $h_0, (h_k^n)_{0\leq k\leq 2n-1}$ are orthonormal. The norm of each function is 1, because clearly $h_0$ has norm 1 and

$$\begin{aligned}\int_0^1 h_k^n(t)^2\,dt &= \int_0^1 2^n\left(\mathbb{1}_{[(2k)2^{-n-1},(2k+1)2^{-n-1}]}(t) - \mathbb{1}_{[(2k+1)2^{-n-1},(2k+2)2^{-n-1}]}(t)\right)^2 dt \\ &= 2^n\left(2^{-n-1} + 2^{-n-1}\right) = 1.\end{aligned}$$

$h_0(t)$ is clearly orthogonal to all other functions, as

$$\int_0^1 h_0(t)h_k^n(t)\,dt = \int_0^1 h_k^n(t)\,dt = 2^{-n/2} - 2^{-n/2} = 0$$

Finally, consider some pair $h_k^n$ and $h_j^m$. Observe that the support of $h_k^n$ is $[k2^{-n}, (k+1)2^{-n})$. If $n = m$ but $k \neq j$, then the functions simply don't overlap at all. Otherwise, without loss of generality $n < m$. Then, the support $[j2^{-m}, (j+1)2^{-m})$ either lies completely inside $[k2^{-n}, (k+1/2)2^{-n})$, completely inside $[(k+1/2)2^{-n}, (k+1)2^{-n})$, or is disjoint from both. In the first case, the integral is 0 because we evaluate $2^{(n+m)/2}\cdot 2^{-m-1} - 2^{(n+m)/2}\cdot 2^{-m-1} = 0$. In the second case, the integral is 0 because we evaluate $-2^{(n+m)/2}\cdot 2^{-m-1} + 2^{(n+m)/2}\cdot 2^{-m-1} = 0$. In the third case, the integral is 0 because the functions don't overlap at all.

**Idea** (Useful Functions) — The functions $g_n^k\colon [0,1] \to [0,\infty)$ are given by

$$g_n^k(t) = \int_0^t h_n^k(s)\,ds.$$

These are important, because they're $\langle h_n^k, [0,t]\rangle$.

Note that the functions $g_n^k$ are continuous and satisfy the following property: For every fixed $n \geq 0$ and $0 \leq k \leq 2^n - 1$, the function $g_n^k$ has support on $[k2^{-n}, (k+1)2^{-n}]$ and is bounded above by $2^{-n/2}$.

We now show the set of (finite) linear combinations of these elements is dense in $L^2([0,1], B([0,1]), dt)$. It is well-known that the simple functions (piecewise constant on intervals) are dense in $L^2$. Any such simple function can be written as a (finite) linear combination of step functions $[0,t]$ (in this problem I will use $[0,t]$ as shorthand for $\mathbb{1}_{[0,t]}$). We'll now show that

$$th_0 + \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} g_k^n(t)h_k^n$$

converges in $L^2$ to $[0,t]$. For all $n \geq 0$, let $i_n = \lfloor 2^n t \rfloor$ (such that $t \in [i_M 2^{-M}, (i_M+1)2^{-M})$). For all $M \geq 0$, consider the function

$$f_M = th_0 + \sum_{n=0}^{M-1} \sum_{k=0}^{2^n-1} g_k^n(t) h_k^n$$

(where the summation is empty if $M = 0$). This function is completely described by

$$f_M(t) = \begin{cases} 1 & \text{if } t \in [0, i_M 2^{-M}) \\ 0 & \text{if } t \in [(i_M+1)2^{-M}, 1) \\ 2^M(t - i_M 2^{-M}) & \text{if } t \in [i_M 2^{-M}, (i_M+1)2^{-M}) \end{cases}$$

This can be proved by an essentially straightforward induction. As a base case, this is clearly true for $M = 0$. On $M \geq 1$, we augment the result for $M-1$ with the $n = M-1$ summation. Observe that $g_k^{M-1}$ vanishes on all $k$ except $k = i_{M-1}$, so our augmentation only affects the third branch of the above expression for $f_M$. There are now two symmetric cases: if $i_M$ is even, then

$$g_{i_{M-1}}^{M-1}(t) = 2^{(M-1)/2}\left(t - i_{M-1}2^{-M+1}\right).$$

One simply adds this to the result for $f_{M-1}$. In the range $[i_M 2^{-M}, (i_M+1)2^{-M}]$, the result is

$$2^{(M-1)/2} \cdot 2^{(M-1)/2}\left(t - i_{M-1}2^{-M+1}\right) + 2^{M-1}(t - i_{M-1}2^{-M+1}) = 2^M(t - i_M 2^{-M}),$$

while on the range $[(i_M+1)2^{-M-1}, (i_M+2)2^{-M}]$,

$$2^{(M-1)/2} \cdot 2^{(M-1)/2}\left(t - i_{M-1}2^{-M+1}\right) - 2^{M-1}(t - i_{M-1}2^{-M+1}) = 0$$

as desired.

Meanwhile, if $i_M$ is odd, then

$$g_{i_{M-1}}^{M-1}(t) = 2^{(M-1)/2}\left((i_{M-1}+1)2^{-M+1} - t\right).$$

This time, in the range $[i_M 2^{-M}, (i_M+1)2^{-M}]$, the result is

$$2^{(M-1)/2} \cdot 2^{(M-1)/2}\left((i_{M-1}+1)2^{-M+1} - t\right) + 2^{M-1}(t - i_{M-1}2^{-M+1}) = 1,$$

while on the range $[(i_M+1)2^{-M-1}, (i_M+2)2^{-M}]$,

$$2^{(M-1)/2} \cdot 2^{(M-1)/2}\left((i_{M-1}+1)2^{-M+1} - t\right) - 2^{M-1}(t - i_{M-1}2^{-M+1}) = 2^M(t - i_M 2^{-M}).$$

Thus, the result is true for $M$.

Observe that the $L^2$ norm of $f_M - [0,t]$ is at most $10 \cdot 2^{-M/2}$, because they differ on an interval of length $2^{-M}$ by at most 2.

**Problem 4.2 —** Suppose that $N_0, (N_n^k)_{0 \leq k \leq 2n-1}$ are independent $N(0,1)$ random variables. Justify the existence of the (unique) Gaussian white noise $G$ on $[0,1]$, with intensity $dt$, such that $G(h_0) = N_0$ and $G(h_k^n) = N_n^k$ for every $n \geq 0$ and $0 \leq k \leq 2n-1$.

This is actually just Hilbert space theory. We already have an orthonormal basis called $h_0, (h_k^n)_{0\le k\le 2n-1}$, and we have a set of independent random variables called $N_0, (N_n^k)_{0\le k\le 2n-1}$, which for the purposes of this section are just vectors inside of $L^2(\Omega, \mathcal{F}, \mathbb{P})$. We can thus define a linear map $G$ from $L^2([0,1], B([0,1]), dt)$ to $L^2(\Omega, \mathcal{F}, \mathbb{P})$ by $G(h_0) = N_0$ and $G(h_k^n) = N_n^k$. There is really nothing else to justify in terms of the existence of this isometry $G$ – this is clearly an isometry, because any element of $L^2([0,1], B([0,1]), dt)$ gets mapped as

$$\sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} a_{n,k}h_k^n \quad \mapsto \quad \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} a_{n,k}N_n^k$$

and both side have norm $\sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} a_{n,k}^2$. The series on the RHS converges in $L^2$ because this summation was finite. This is just not very deep.

The point is that this is actually a Gaussian white noise, i.e. all elements of the image actually live inside the Gaussian space $H$ which is the $L^2$-closure of $\operatorname{span}(N_0, (N_n^k)_{0\le k\le 2n-1})$. This is clear, because you can literally look at the mapping above; its some $L^2$-convergent series (where each prefix is a Gaussian random variable) and thus the limit is a Gaussian random variable.

**Problem 4.3** — For every $t \in [0,1]$, set $B_t = G([0,t])$. Verify that

$$B_t = tN_0 + \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} g_n^k(t)N_n^k,$$

where the series converges in $L^2$.

Fortunately, we already did the hard work to show that

$$[0,t] = th_0 + \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} g_n^k(t)h_n^k.$$

By elementary properties of Hilbert spaces,

$$G([0,t]) = tG(h_0) + \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} g_n^k(t)G(h_n^k) = tN_0 + \sum_{n=0}^{\infty}\sum_{k=0}^{2^n-1} g_n^k(t)N_n^k.$$

**Problem 4.4** — For every integer $m \ge 0$ and every $t \in [0,1]$ set

$$B_t^{(m)} = tN_0 + \sum_{n=0}^{\infty}\left(\underbrace{\sum_{k=0}^{2^n-1} g_n^k(t)N_n^k}_{C_t^{(m,n)}}\right)$$

Verify that the continuous functions $t \mapsto B_t^{(m)}(\omega)$ converge uniformly on $[0,1]$ as $m \to \infty$, for a.a. $\omega \in \Omega$.

**Idea** (Hint) — If $N$ is $N(0,1)$-distributed, prove the bound $\mathbb{P}(|N| > a) \le e^{-a^2/2}$ for $a \ge 0$, and use this estimate to bound the probability of the event $\{\sup_{0\le k\le 2^n-1} \left|N_n^k\right| > 2^{n/4}\}$, for every fixed $n \ge 0$.

Sure, let's follow the hint. Here's a fun trick:

$$\begin{aligned}
I &= \int_{|N|<a} \frac{1}{\sqrt{2\pi}} e^{-N^2/2}\, dN \\
I^2 &= \int_{|x|<a,|y|<a} \frac{1}{2\pi} e^{-x^2/2} e^{-y^2/2}\, dx\, dy \\
&\ge \int_{r<a} r e^{-r^2/2}\, dr \\
&= \int_0^{a^2/2} e^{-s}\, ds = 1 - e^{-a^2/2} \\
\implies I &\ge \sqrt{1 - e^{-a^2/2}} \\
\implies \mathbb{P}(|N| > a) &\le 1 - \sqrt{1 - e^{-a^2/2}} \le e^{-a^2/2}.
\end{aligned}$$

With this in mind, the probability of the event $A_n = \{\sup_{0\le k\le 2^n-1} \left|N_n^k\right| > 2^{n/4}\}$ is at most $2^n$ times the probability that $|N| > 2^{n/4}$, so $2^n \cdot \exp\left(-2^{n/2}/2\right)$. This is clearly summable, thus by Borel-Cantelli, the events $(A_n)_n$ almost surely occur only finitely often. We now claim that only finitely many of these events occur, $B_t^{(m)}$ converges uniformly on $[0,1]$. By truncating the events where it does occur, it suffices to prove this statement in the case that none of the $A_n$ occur.

But this is completely obvious, because each of the $g_n^k$ is bounded above by $2^{-n/2}$, and thus each inner summation $C_t^{(m,n)}$ in the series is uniformly bounded by $2^{n/4} \cdot 2^{n/2} = 2^{-n/4}$. The series $2^{-n/4}$ are (absolutely) summable, thus by the Weierstrass M-test, the series converges uniformly on $[0,1]$.

**Idea** (Diagram from Book) —

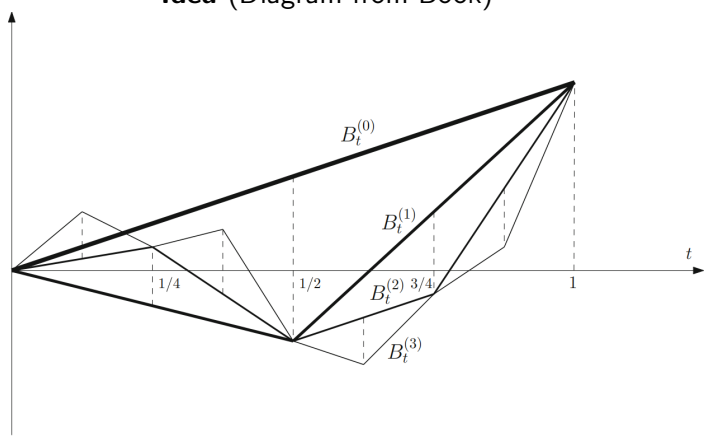


**Fig. 1.1** Illustration of the construction of $B_t^{(m)}$ in Exercise 1.18, for $m = 0, 1, 2, 3$. For the clarity of the figure, lines become thinner when $m$ increases. The lengths of the dashed segments are determined by the values of $N_0$ and $N_k^m$ for $m = 0, 1, 2$

**Problem 4.5 —** Conclude that we can, for every $t \in [0, 1]$, select a random variable $B'_t$ which is a.s. equal to $B_t$, in such a way that the mapping $t \mapsto B'_t(\omega)$ is continuous for every $\omega \in \Omega$.

A uniformly convergent sequence of continuous functions converges to a continuous function. Thus, it is already almost surely true that $B_t(\omega)$ is continuous, and we can set $B' = B$ in those cases. In the (measure 0) event that it is not, simply define $B'_t(\omega) = 0$. This is a.s. equal to $B_t$, and now the mapping $t \mapsto B_t(\omega)$ is continuous for every $\omega \in \Omega$.